

AN ASYMPTOTIC REPRESENTATION OF THE DISTRIBUTION OF A SPECIFIC QUADRATIC FORM

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Summary: There exists a variety of series expansions for the distribution of a quadratic form in a normal vector. The convergence of each of the series expansions is largely determined by the structure of the eigenvalues of the quadratic form. A series expansion for the distribution of a positive definite quadratic form in a normal vector for the central case is derived. The series converges rapidly and is also simple to compute. The first three terms of the series are suggested as an approximation to the distribution function. An application is presented.

1. INTRODUCTION

Consider a positive definite quadratic form in a normal vector for the central case:

$$(1.1) \quad \xi = \mathbf{x}' \mathbf{A} \mathbf{x}$$

The random vector $\underline{x} : p \times 1$ has a standard normal distribution and $A : p \times p$ is a symmetric positive definite matrix.

Johnson and Kotz (1970, pp. 149-168) discuss a variety of series expansions for the distribution of $\underline{s}$. The convergence of each of the series expansions is largely determined by the structure of the eigenvalues of A . The magnitudes of the eigenvalues also determine the choice of arbitrary constants in the series expansions.

The distribution of $\underline{s} = \underline{x}'A\underline{x}$ is the same as that of $\underline{s} = \underline{x}'\Delta\underline{x}$, where $\Delta = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_p)$ and $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p$ are the eigenvalues of A . In paragraph 2 a series expansion for the distribution of $\underline{s}$, for the case when the λ_j 's have a specific structure, is derived.

When each λ_j has the form

$$(1.2) \quad \lambda_j^{-1} = gn + d_j \quad j=1,2,\dots,p$$

with n large and $g \neq 0$ and d_j constants, independent of n , it is shown that the distribution function of $\underline{s}$ can be approximated with a weighted sum of that of the logarithm of a beta random variable, with specified parameters, and two chi-square distribution functions.

In paragraph 3 an application is discussed.

2. THE SERIES EXPANSION

The characteristic function of $\underline{s} = \underline{z}\underline{x}'A\underline{x}$, where $\underline{z}$ is an arbitrary constant, is

$$(2.1) \quad \phi(t) = |I - 2it\zeta A|^{-\frac{1}{2}}$$

$$(2.2) \quad = \theta^{\frac{1}{2}p} \exp \left[\sum_{r=1}^{\infty} \kappa_r (\theta^r - 1) \right]$$

$$(2.3) \quad = \theta^{\frac{1}{2}p} |\zeta A|^{-\frac{1}{2}} \exp \left[\sum_{r=1}^{\infty} \kappa_r \theta^r \right]$$

$$(2.4) \quad = \theta^{\frac{1}{2}p} \sum_{k=0}^{\infty} \eta_k \theta^k$$

where

$$(2.5) \quad \theta = \theta(t) = (1 - 2it)^{-1}$$

$$(2.6) \quad \eta_0 = |\zeta A|^{-\frac{1}{2}}$$

$$\eta_k = \frac{1}{k} \sum_{r=1}^k r \kappa_r \eta_{k-r} \quad k=1,2,3,\dots$$

$$(2.7) \quad \kappa_r = \frac{1}{2r\zeta^r} \operatorname{tr}(\zeta I - A^{-1})^r \quad r=1,2,3,\dots$$

(See, e.g. Johnson and Kotz (1970), pp. 156-158.)

The choice for ζ , which minimizes κ_r , is

$$(2.8) \quad \zeta = \frac{1}{p} \operatorname{tr}(A^{-1}) = \frac{1}{p} \sum_{j=1}^p \lambda_j^{-1} = g_n + \bar{d}$$

With this choice, we have

$$(2.9) \quad \kappa_1 = \eta_1 = 0$$

$$\kappa_r = \frac{1}{2r\zeta^r} \sum_{j=1}^p (\bar{d} - d_j)^r \quad r=2,3,4,\dots$$

Hence

$$(2.10) \quad \kappa_r = O(n^{-r}) \quad \text{and} \quad \eta_r = O(n^{-r}) \quad r=2,3,4,\dots$$

Using the Maclaurin expansion for $\exp(x)$ in (2.2) and grouping

terms of $O(n^{-5})$ together, we have

$$(2.11) \quad \phi(t) = \theta^{\frac{1}{2}p} \left[\sum_{j=0}^4 \eta_j^* \theta^j + R_5(\theta) \right]$$

where

$$(2.12) \quad \eta_0^* = 1 - \eta_2^* - \eta_3^* - \eta_4^*$$

$$\eta_1^* = 0$$

$$\eta_2^* = \kappa_2 - \kappa_2^2$$

$$\eta_3^* = \kappa_3$$

$$\eta_4^* = \kappa_4 + \frac{1}{2}\kappa_2^2$$

and

$$(2.13) \quad R_5(\theta) = \sum_{j=0}^4 (\eta_j - \eta_j^*) \theta^j + \sum_{j=0}^{\infty} \eta_j \theta^j \\ = O(n^{-5}).$$

It is important to note that if the λ_j^{-1} are symmetric about ζ , then κ_r and hence η_r and η_r^* will be zero for odd r .

If $A = \lambda I$, then $\eta_k = 0$ for all nonzero k , and $\eta_0 = 1$, hence $\phi(t) = \theta^{\frac{1}{2}p}$, which is the $\chi^2(p)$ characteristic function.

It will now be shown that, for $p > 2$,

$$(2.14) \quad \phi(t) = a\phi_{\chi}(t) + b\theta^{\frac{1}{2}p} + \kappa_3\theta^{\frac{1}{2}p+3} + O(n^{-5})$$

where $\phi_{\chi}(t)$ is the characteristic function of $-2\zeta_0 \ln \chi$, with χ having a specified beta distribution. The arbitrary constants a , b and ζ_0 will be chosen in such a way that $a + b + \kappa_3 = 1$ and ζ_0 will depend on n .

Let $y = -2\zeta_0 \ln x$, where $y \sim B(\alpha, \beta)$. The characteristic function of v is:

$$\begin{aligned} (2.15) \quad \phi_{\tilde{y}}(t) &= \frac{\Gamma(\alpha+\beta) \cdot \Gamma(\alpha-2\zeta_0 it)}{\Gamma(\alpha) \cdot \Gamma(\alpha+\beta-2\zeta_0 it)} \\ &= \frac{\Gamma(\alpha+\beta) \cdot \Gamma(\alpha-\zeta_0+\zeta_0(1-2it))}{\Gamma(\alpha) \cdot \Gamma(\alpha+\beta-\zeta_0+\zeta_0(1-2it))}. \end{aligned}$$

Erdélyi (1953, p. 48) gives the following series expansion, which is asymptotic in x and valid for $|\frac{h-1}{x}| < 1$ and $R_e(x) > 0$.

($R_e(x)$ denotes the real part of x .)

$$(2.16) \quad \ln \Gamma(x+h) = \frac{1}{2} \ln(2\pi) + (x+h-\frac{1}{2}) \ln(x) - x + \sum_{r=1}^{\infty} \frac{(-1)^{r+1}}{r(r+1)x^r} B_{r+1}(h)$$

where $B_r(h)$ is a Bernoulli polynomial.

Hence

$$(2.17) \quad \ln \Gamma(x+h_1) - \ln \Gamma(x+h_2) = (h_1 - h_2) \ln x + \sum_{r=1}^{\infty} w_r x^{-r}$$

and

$$\begin{aligned} (2.18) \quad \Gamma(x+h_1)/\Gamma(x+h_2) &= x^{h_1-h_2} e^{\sum_{r=1}^{\infty} w_r x^{-r}} \\ &= x^{h_1-h_2} \sum_{k=0}^{\infty} \psi_k x^{-k} \end{aligned}$$

$$(2.19) \quad w_r = \frac{(-1)^{r+1}}{r(r+1)} [B_{r+1}(h_1) - B_{r+1}(h_2)]$$

and

$$(2.20) \quad \psi_0 = 1$$

$$\psi_k = \frac{1}{k} \sum_{r=0}^{k-1} (k-r) w_{k-r} \psi_r.$$

Putting $x = \zeta_0(1-2it)$; $h_1 = \alpha - \zeta_0$ and $h_2 = \alpha + \beta - \zeta_0$ in (2.18), (2.20) yields a series expansion for $\phi_{\sim}(t)$:

$$(2.21) \quad \phi_{\sim}(t) = \theta^{\beta} \exp \left[\sum_{r=1}^{\infty} w_r (\theta^r - 1) \right]$$

$$(2.22) \quad = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\zeta_0^{\beta}} \theta^{\beta} \exp \left[\sum_{r=1}^{\infty} w_r \theta^r \right]$$

$$(2.23) \quad = \theta^{\beta} \sum_{k=0}^{\infty} \psi_k \theta^k$$

(with $\theta = \theta(t) = (1-2it)^{-1}$ as before)

where

$$(2.24) \quad w_r = \frac{(-1)^{r+1}}{r(r+1)\zeta_0^r} \left[B_{r+1}(\alpha-\zeta_0) - B_{r+1}(\alpha+\beta-\zeta_0) \right].$$

Because (see e.g. Abramowitz and Stegun (1965, pp. 804-806))

$$(2.25) \quad B_r(1-x) = (-1)^r B_r(x) \quad r=1,2,3,\dots$$

it follows that if α is chosen in such a way that

$$(2.26) \quad \alpha - \zeta_0 = 1 - (\alpha + \beta - \zeta_0) \\ \therefore \alpha = \zeta_0 - \frac{1}{2}(\beta - 1)$$

then

$$(2.27) \quad w_r = \frac{1}{r(r+1)\zeta_0^r} \left[2B_{r+1}\left(\frac{1}{2}(\beta+1)\right) \right] \\ = 0 \quad \text{for } r \text{ odd}$$

and

$$(2.28) \quad \psi_0 = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\zeta_0^\beta}$$

$$\psi_k = \frac{1}{k} \sum_{r=0}^{k-1} (k-r)w_{k-r}\psi_r$$

$$= 0 \quad \text{for } k \text{ odd.}$$

Because ζ_0 will be chosen to depend on n , we will have

$$(2.29) \quad w_j = O(n^{-j}) \quad \text{and hence } \psi_j = O(n^{-j}) \quad \text{for } j=2,4,6,\dots$$

Expanding (2.21), using arguments similar to those used in obtaining (2.11), yields

$$(2.30) \quad \phi_{\sim}(t) = \theta^\beta \left[\sum_{j=0,2,4} \psi_j^* \theta^j + Q_6(\theta) \right]$$

where

$$(2.31) \quad \psi_0^* = 1 - \psi_2^* - \psi_4^*$$

$$\psi_2^* = w_2 - w_2^2$$

$$\psi_4^* = w_4 + \frac{1}{2}w_2^2$$

$$(2.32) \quad Q_6(\theta) = \sum_{j=0,2,4} (\eta_j - \eta_j^*) \theta^j + \sum_{j=6,8,10} \eta_j \theta^j$$

$$= O(n^{-6}).$$

Hence, from (2.11) and (2.30):

$$(2.33) \quad \phi(t) - a\phi_{\sim}(t) = \theta^{\frac{1}{2}P} \sum_{j=0}^4 \eta_j^* \theta^j - a\theta^\beta \sum_{j=0}^4 \psi_j^* \theta^j + \theta^{\frac{1}{2}P} R_5(\theta) - a\theta^\beta Q_6(\theta).$$

By requiring the coefficients of the same powers of θ to be of the same order of n in the RHS of (2.33), it follows that

$$(2.34) \quad \beta = \frac{1}{2}p.$$

Hence

$$(2.35) \quad \phi(t) - a\phi_{\sim}(t) = \sum_{k=0}^4 \mu_k \theta^{\frac{1}{2}p+k} + O(n^{-5})$$

where

$$(2.36) \quad \mu_k = \eta_k^* - a\psi_k^* \quad k=0,1,2,3,4.$$

If the λ_j^{-1} are symmetric about ζ , then $\mu_r = 0$ for odd r and the error on the RHS of (2.35) will be $O(n^{-6})$. The constants a and ζ_0 may now be selected in such a way as to make the coefficients of $\theta^{\frac{1}{2}p+2}$ and $\theta^{\frac{1}{2}p+4}$, that is, μ_2 and μ_4 , zero. Hence

$$(2.37) \quad a = \frac{\eta_2^*}{\psi_2^*} = \frac{\eta_4^*}{\psi_4^*} = \frac{\kappa_2 - \kappa_2^2}{w_2 - w_2^2} = \frac{\kappa_4 + \frac{1}{2}\kappa_2^2}{w_4 + \frac{1}{2}w_2^2}.$$

Because

$$(2.38) \quad \kappa_2 = \frac{1}{4\zeta^2} \sum (\bar{d} - d_j)^2 = \kappa_2 / \zeta^2 \quad (\text{say})$$

$$\kappa_4 = \frac{1}{8\zeta^4} \sum (\bar{d} - d_j)^4 = \kappa_4 / \zeta^4 \quad (\text{say})$$

and, using Abramowitz and Stegun (1965, p. 809),

$$(2.39) \quad w_2 = \frac{p(p^2 - 4)}{192\zeta_0^2}$$

$$w_4 = \frac{p(p^2 - 4)(3p^2 - 28)}{30720\zeta_0^4} = w_2 \frac{(3p^2 - 28)}{160\zeta_0^2}.$$

It follows that μ_2 and μ_4 will be zero when

$$(2.40) \quad \zeta_0^2 = \delta\zeta^2 + \gamma$$

where

$$(2.41) \quad \delta = \frac{192k_2}{p(p^2-4)} \left[\frac{\frac{p(p^2-4)(3p^2-28)}{30720} + \frac{p^2(p^2-4)^2}{2(192)^2}}{k_4 + \frac{1}{2}k_2^2} \right]$$

$$= \frac{k_2(5p-14)(p+6)(p+4)}{960[2k_4 + k_2^2]}$$

$$(2.42) \quad \gamma = \frac{p(p^2-4)}{192} - \delta \cdot k_2$$

$$(2.43) \quad a = \frac{192k_2}{p(p^2-4)\delta} \cdot \frac{\zeta_0^4}{\zeta^4}.$$

Hence δ , γ and a are all independent of n . These choices for ζ_0 , α and β ensure the validity of the expansion (2.30) because they satisfy the conditions for the validity of (2.16). Note that the solution only exists if $p > 2$. The characteristic function $\phi(t)$ now becomes

$$(2.44) \quad \phi(t) = a\phi_{\gamma}(t) + (1-a-\kappa_3)\theta^{\frac{1}{2}p} + \kappa_3\theta^{\frac{1}{2}p+3} + O(n^{-5})$$

which is in the form (2.14), with $b = 1 - a - \kappa_3$. Taking the inverse transform of each term, the distribution function of ζ_{γ} may be expressed as

$$(2.45) \quad F(\xi) = aF_{\gamma}(\xi) + bG_p(\xi) + \kappa_3G_{p+6}(\xi) + O(n^{-5}).$$

$F_{\gamma}(\cdot)$ corresponds to $\phi_{\gamma}(t)$ and $G_p(\cdot)$ is the distribution function of $x^2(p)$.

Hence the distribution function of $\underline{z} = \underline{x}'A\underline{x}$ is

$$(2.46) \quad F_{\underline{z}}(s) = a \cdot F_{\underline{z}}(s) + b \cdot G_p(\zeta s) + \kappa_3 G_{p+6}(\zeta s) + O(n^{-5}).$$

$F_{\tilde{z}}(\cdot)$ is the distribution function of $\tilde{z} = -2(\zeta_0/\zeta)\ln \chi$, where $\chi \sim B(\alpha, \frac{1}{2}p)$. If the λ_j^{-1} are symmetric about ζ , then $\kappa_3 = 0$ and the error term is $O(n^{-6})$. If $A = \lambda I$, a , κ_3 and the error term are all zero, and $\zeta = 1$. Hence $F_{\tilde{s}}(s) = G_p(s)$. Generally, the first three terms on the RHS of (2.46) yield a good approximation to the distribution function of $\tilde{s}$.

Note that the exact distribution function of $\tilde{s}$ is:

$$(2.47) \quad F_{\tilde{s}}(s) = aF_{\tilde{z}}(s) + \sum_{k=0}^{\infty} (\eta_k - a\psi_k) G_{p+2k}(\zeta s).$$

It has been shown that, with the indicated choices for all the constants,

$$(2.48) \quad \begin{aligned} \eta_0 - a\psi_0 &= b + O(n^{-5}) \\ \eta_1 - a\psi_1 &= 0 \\ \eta_2 - a\psi_2 &= O(n^{-5}) \\ \eta_3 - a\psi_3 &= \kappa_3 + O(n^{-5}) \\ \eta_4 - a\psi_4 &= O(n^{-5}). \end{aligned}$$

3. APPLICATION: WILKS' LIKELIHOOD RATIO CRITERION

Consider the statistic

$$(3.1) \quad \underline{u} = \prod_{j=1}^p \tilde{z}_j$$

where $\tilde{z}_j$ are independently $B(\frac{n+1-j}{2}, \frac{m}{2})$ distributed.

The characteristic function of $-2\ln \underline{u}$ is (see, e.g. Anderson (1958), p. 207)

$$(3.2) \quad \phi(t) = \prod_{j=1}^p \frac{\Gamma(\frac{1}{2}(n+1-j) - 2it) \cdot \Gamma(\frac{1}{2}(n+m+1-j))}{\Gamma(\frac{1}{2}(n+1-j)) \cdot \Gamma(\frac{1}{2}(n+m+1-j) - 2it)}.$$

Using the fact that (for $\gamma = 1, 2, 3, \dots$)

$$(3.3) \quad \frac{\Gamma(\alpha+\gamma)}{\Gamma(\alpha+\gamma-2it)} = \left[1 - \frac{2it}{\alpha-\gamma-1}\right]^{-1} \cdot \left[1 - \frac{2it}{\alpha+\gamma-2}\right]^{-1} \dots \left[1 - \frac{2it}{\alpha}\right]^{-1} \cdot \frac{\Gamma\alpha}{\Gamma(\alpha-2it)}$$

we may write (3.2), for even m ,

$$(3.4) \quad \phi(t) = \prod_{j=1}^p |I - 2itA_j|^{-\frac{1}{2}} = |I - 2itA|^{-\frac{1}{2}}$$

where

$$(3.5) \quad A_j : m \times m = \text{diag}\left(\frac{2}{n+1-j}, \frac{2}{n+1-j}, \frac{2}{n+3-j}, \frac{2}{n+3-j}, \dots, \frac{2}{n+m-1-j}, \frac{2}{n+m-1-j}\right)$$

and

$$(3.6) \quad A : mp \times mp = \text{Diag}(A_1, A_2, \dots, A_p).$$

$$(3.7) \quad \zeta = \frac{1}{mp} \text{tr}(A^{-1}) = \frac{1}{2}n + \frac{1}{2}(m-p-1).$$

Now (3.4) is in the form (2.1). The eigenvalues of A are in the form (1.2) and symmetric about ζ . Also

$$(3.8) \quad \kappa_r = \frac{1}{r\zeta^r} \sum_{j=1}^p \sum_{k=1}^{\frac{1}{2}m} \{ \frac{1}{2}(j - \frac{1}{2}(p+1)) - (k - \frac{1}{2}(m+2)) \}^r$$

$$= 0 \quad r \text{ odd.}$$

Hence, (see (2.38)):

$$(3.9) \quad k_2 = \frac{1}{192} mp(p^2 + m^2 - 5)$$

$$k_4 = \frac{1}{8.3840} (3p^4 + 10p^2m^2 + 3m^4 - 50m^2 - 50p^2 + 159)$$

From (2.39):

$$(3.10) \quad w_2 = \frac{1}{192\zeta_0^2} mp(m^2 p^2 - 4)$$

$$w_4 = \frac{1}{160\zeta_0^2} (3m^2 p^2 - 28) \cdot w_2.$$

It follows that

$$(3.11) \quad \zeta_0^2 = \delta \zeta^2 + \gamma$$

$$\delta = \frac{k_2(5mp - 14)(mp + 6)(mp + 4)}{960[2k_4 + k_2^2]}$$

$$\gamma = \frac{1}{192} mp(m^2 p^2 - 4) - \delta \cdot \kappa_2$$

$$a = \frac{p^2 + m^2 - 5}{m^2 p^2 - 4} \cdot \frac{\zeta_0^2}{\delta \cdot \zeta^4}.$$

Also:

$$(3.12) \quad \beta = \frac{1}{2} mp$$

$$\alpha = \zeta_0 - \frac{1}{2}(\beta - 1).$$

From (2.46), the distribution function of $-2\ln \underline{u}$ is given by

$$(3.13) \quad F(\xi) = aF_{\underline{z}}(\xi) + bG_p(\zeta\xi) + \kappa_3 G_{p+6}(\zeta\xi) + O(n^{-6}).$$

Hence the distribution function of $\underline{u}$ is given by

$$(3.14) \quad F_{\underline{u}}(u) = aF_{\underline{y}}(u^{\zeta_0/\zeta}) + bG_p(-2\zeta \ln u) + \kappa_3 G_{p+6}(-2\zeta \ln u) + O(n^{-6})$$

where y is distributed $B(\alpha, \frac{1}{2}mp)$. Note that if $p=1$, then $\delta=1$, $\gamma=0$, $\zeta_0=\zeta$ and $a=1$. Also, if either $p=2$ or $m=2$, then $\delta=4$, $\gamma=0$, $\zeta_0=2\zeta$ and $a=1$, which agrees with the fact that, in this case, $\underline{u}$ is exactly distributed as the square of a beta random variable. The expansion (3.12) was derived for even m only. Using a technique similar to that derived in Crowther (1980)

it can be proved that the expansion is also valid for odd m .

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